

REVIEW

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Human–AI hybrid finance: from AI tools to decision systems

Gang Kou¹, Yan Li², Haomin Wang³ and Xin Wang^{1*} 

*Correspondence:
wxin@swufe.edu.cn

¹ School of Business Administration, Faculty of Business Administration, Southwestern University of Finance and Economics, No. 555 Liutai Avenue, Wenjiang District, Chengdu 611130, Sichuan, People's Republic of China

² School of Accounting, Faculty of Accounting, Southwestern University of Finance and Economics, No. 555 Liutai Avenue, Wenjiang District, Chengdu 611130, Sichuan, People's Republic of China

³ School of Management Science and Engineering, Southwestern University of Finance and Economics, No. 555 Liutai Avenue, Wenjiang District, Chengdu 611130, Sichuan, People's Republic of China

Abstract

Artificial intelligence (AI) in finance is commonly reviewed by method, data type, or application domain. These perspectives are essential, but they understate a deeper shift: AI is moving from a predictive tool to a component of human–AI hybrid financial decision systems. This integrative and conceptual review synthesizes literature across finance, management, human–computer interaction (HCI), and AI to examine how humans and AI jointly participate in information acquisition, prediction, recommendation, approval, execution, monitoring, and learning. We argue that the central question is moving from model performance to decision architecture: how authority, oversight, and accountability should be allocated across financial workflows. We show that human–AI complementarity in finance is conditional rather than automatic, depending on task structure, private information, feedback quality, incentives, explanation design, and governance. We also argue that AI-mediated financial decisions are reflexive: they reshape organizational workflows, prices, liquidity, credit allocation, and the future data on which subsequent decisions rely. The review integrates evidence on methods, data, scenarios, explainability, trust, governance, financial large language models (FinLLMs), and agentic finance, and organizes the field around an integrated decision-system framework consisting of five connected constructs—delegation frontier, reliance wedge, decision-useful explainable artificial intelligence (XAI), meaningful oversight, and reflexive AI loop—to support cumulative research on investment, trading, credit, asset management, risk, compliance, and financial regulation.

Keywords: Artificial intelligence in finance, Human–AI hybrid finance, Human–AI collaboration, FinLLMs, Explainable AI, Financial decision-making, Governance, Agentic finance

Introduction: from AI in finance to human–AI hybrid finance

Finance is a field of decisions under uncertainty. Data become economically meaningful when they change a decision: whether to lend, trade, hedge, rebalance, approve, invest, disclose, intervene, or regulate. For this reason, the spread of artificial intelligence (AI) in finance should not be understood only as the adoption of new predictive models. It is also a transformation in the architecture of financial decision-making. AI systems now acquire information from structured and unstructured data, convert noisy observations into signals, generate recommendations, trigger alerts, execute actions, and learn

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from outcomes. The relevant unit of analysis is therefore moving from the *model* to the *human–AI hybrid financial decision system*: a sociotechnical arrangement in which humans, algorithms, organizations, markets, and regulators jointly produce financial outcomes.

This shift builds on a long literature. Early financial prediction and scoring research focused on ratios, default classification, and econometric decision rules (Beaver 1966; Ohlson 1980). More recent surveys show how AI in finance expanded into supervised learning, deep learning, reinforcement learning, and large language models (LLMs), including financial large language models (FinLLMs), across asset pricing, trading, credit, fraud detection, accounting, and risk management (Cao 2022; Goodell et al. 2021; Ahmed et al. 2022; Bahoo et al. 2024; Cao et al. 2024; Goldstein et al. 2021; Gu et al. 2020; Henrique et al. 2019; Hambly et al. 2023; Li et al. 2023b; Lee et al. 2025). Finance is also one of the domains most exposed to generative AI (GenAI) because it is information-intensive, text heavy, highly codified, and operationally dependent on rapid judgment under uncertainty (Eisfeldt and Schubert 2025; Wu et al. 2023; Yang et al. 2023; Xie et al. 2024; Li et al. 2025).

Existing reviews are indispensable, but their common organization also reflects their primary object of analysis. Method-, data-, and application-oriented reviews mainly ask what AI systems can process, predict, classify, optimize, or automate in financial settings, and how well they perform on those tasks. This review uses that literature as a foundation, but it is not intended to be a comprehensive technical survey of all financial AI methods. Its purpose is instead to provide a decision-system-oriented synthesis. We ask how AI capabilities move from analytical tools into financial decision systems, and how their use changes the allocation of information, authority, reliance, explanation, oversight, and accountability among humans, algorithms, organizations, and regulators.

As AI systems have expanded from predictive models into language systems, recommender systems, tool-using agents, and workflow components, AI is no longer only an analytical tool used before decisions are made. It increasingly becomes part of the infrastructure through which financial decisions are produced, justified, executed, monitored, and revised. Put differently, AI changes the structure of judgment: who sees what, who interprets what, who can override whom, and who bears responsibility when outcomes fail. A credit score matters because it affects approval, pricing, monitoring, fairness, and accountability. A return forecast matters because it changes positions, risk limits, and client communication. A FinLLM matters not only because it summarizes a filing, but because it may redirect analyst attention, draft advice, generate compliance language, call tools, and eventually act through agents.

We therefore define *human–AI hybrid finance* as financial decision systems in which humans and AI jointly participate in one or more stages of information acquisition, interpretation, prediction, recommendation, approval, override, execution, monitoring, audit, and learning. This definition is broader than the familiar phrase “human in the loop.” In many financial workflows, the human is not simply inserted into an automated pipeline; rather, AI is inserted into human, organizational, regulatory, and market processes. The human–AI hybrid finance lens matters because finance is high-stakes, many tasks combine forecasting with interpretation and escalation, human–AI complementarity is conditional rather than automatic, and FinLLMs and agents are turning AI into a

communication and execution layer (Vaccaro et al. 2024; Fügener et al. 2022; Dietvorst et al. 2015; Logg et al. 2019).

We organize this review around one motivating puzzle: strong stand-alone AI does not automatically produce strong hybrid financial decisions. This puzzle matters because it reveals a limit of model-centered evaluation. Once AI is embedded in financial workflows, outcomes are not determined by model performance alone. They also depend on who has authority to act, how humans rely on AI, whether model outputs can be explained and contested, whether human oversight is substantive, and how AI-mediated actions reshape future data, workflows, and markets.

Answering this puzzle requires repositioning AI within the financial decision system. AI may enter a workflow as a screening device, recommender, explanation generator, monitoring system, bounded executor, or learning component. In each role, it changes a different part of the decision architecture. It can shift decision rights, alter human reliance, produce decision-relevant evidence, change the meaning of oversight, and create feedback loops that reshape the environment being analyzed. These five entry points motivate the integrated decision-system framework developed in this review (Table 1).

The framework is intentionally organized around decision-system functions rather than around model families. It explains why hybrid decision performance depends not only on what AI predicts, but also on how authority, reliance, explanation, oversight, and feedback are designed. Taken together, these constructs connect workflow design, behavioral calibration, explanation, oversight, and system feedback. They also provide the vocabulary through which the rest of the review compares otherwise diverse settings such as lending, trading, portfolio management, advisory work, compliance, insurance, and financial supervision.

Finance is a distinctive setting for human–AI hybrid financial decision systems for three reasons. First, financial decisions are consequential and often distributive: credit approval, insurance pricing, investment advice, and fraud escalation allocate opportunities, costs, and scrutiny across people and firms. Second, financial decisions are deeply

Table 1 An integrated decision-system framework for human–AI hybrid finance

Framework component	Decision-system function	Compact definition
Delegation frontier	Authority allocation	The stage-specific boundary at which decision authority shifts between humans and AI across information acquisition, recommendation, approval, execution, monitoring, and learning
Reliance wedge	Reliance calibration	The gap between actual reliance on AI and warranted reliance given relative accuracy, uncertainty, incentives, and accountability
Decision-useful explainable artificial intelligence (XAI)	Justification and contestability	Explanation that improves actionability, auditability, contestability, and accountability in financial decisions, rather than merely increasing perceived transparency
Meaningful oversight	Substantive human control	Human supervision that has sufficient information, authority, time, independence, and responsibility to affect outcomes
Reflexive AI loop	Feedback and system adaptation	The process through which AI-mediated decisions reshape organizational routines, customer behavior, market conditions, and the future data on which later decisions rely

embedded in regulation and organizational accountability; what matters is not only a model output, but the documented decision, who approved it, how exceptions were handled, and whether the decision can be explained and contested. Third, finance is strategic and feedback-intensive: borrowers, traders, firms, insurers, and fraudsters react to models, while markets aggregate model-mediated actions and generate missing counterfactuals. For that reason, evidence on financial AI should distinguish among three claims that are often conflated: a *technical claim* about signal extraction, an *organizational claim* about workflow performance, and a *market/welfare claim* about externalities, stability, and long-run outcomes (Fig. 1).

Against this background, the contribution of this review is threefold. First, it reframes AI in finance from a collection of predictive tools to a problem of human–AI hybrid financial decision systems. This shifts the unit of analysis from the isolated model to the workflow in which humans and AI jointly acquire information, interpret signals, recommend actions, approve or override decisions, execute tasks, monitor outcomes, and learn from feedback. Second, it integrates four streams of literature that are often studied separately: AI methods in finance, human–AI collaboration and decision support, explainable and trustworthy AI, and generative or agentic AI in finance. Third, it develops an integrated decision-system framework for human–AI hybrid finance. The framework is organized around five connected constructs—the delegation frontier, reliance wedge, decision-useful XAI, meaningful oversight, and reflexive AI loop—which together specify how AI enters financial decision systems through authority allocation, reliance calibration, explanation and justification, substantive oversight, and feedback from AI-mediated action.

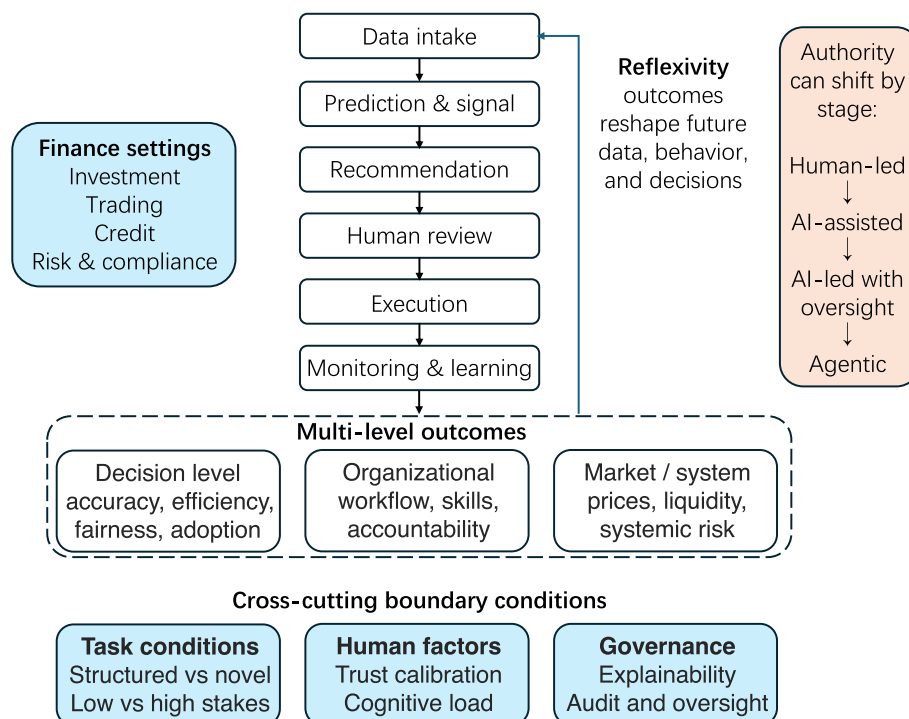


Fig. 1 A decision-system framework for human–AI hybrid finance

Review scope and approach

This article is an integrative and conceptual review rather than a systematic review or a comprehensive technical survey of financial AI applications. Its purpose is not to catalogue every AI method or every use case in finance, but to synthesize representative, foundational, and recent studies that help explain how AI's role in financial decision-making has evolved from a stand-alone analytical tool to a component of human–AI hybrid financial decision systems. We selected and organized the literature according to its relevance to this central question. AI-in-finance studies are used to trace the evolution of methods, data objects, and application scenarios. Human–AI collaboration and human–computer interaction (HCI) research is included to explain reliance, delegation, override, and decision support. Explainable AI, trustworthy AI, and governance research is included to examine transparency, accountability, oversight, and contestability. FinLLM, generative AI, and agentic AI studies are included because they show how AI is becoming a language, workflow, and action layer in finance. Financial regulation and systemic-risk work is included where it helps connect AI-mediated decisions to institutional governance, market feedback, and broader financial stability. The literature is therefore organized around the development of a decision-system framework rather than around an exhaustive inventory of technical methods.

“[The rise of AI: tools and applications in finance](#)” section maps the rise of AI in finance by method, data object, and scenario. “[AI's challenge: when human–AI collaboration works, and when it fails](#)” section asks when human–AI collaboration improves decisions and when it fails. “[Making human–AI hybrid finance work: explainability, trust, and governance](#)” section examines explainability, trust, and governance. “[The future of human–AI hybrid finance](#)” and “[Conclusion: from prediction to delegation](#)” sections turn to future decision architectures, workflow redesign, agents, feedback loops, and a compact research agenda. The aim is not to review every model in isolation, but to explain how finance may be moving from AI-assisted tasks toward human–AI hybrid financial decision systems in which prediction, delegation, explanation, and responsibility are jointly designed.

The rise of AI: tools and applications in finance

AI became influential in finance because methods, data, and use cases evolved together. Financial institutions generate large quantities of labeled and semi-labeled data; markets provide frequent feedback; many tasks are repeated at scale; and financial decisions are economically measurable. At the same time, the growth of unstructured and alternative data created problems that could not be addressed well by traditional linear models alone. The result was an expansion from tabular prediction to representation learning, sequential control, language-based analysis, and increasingly agentic workflows. Existing reviews map this expansion from different angles: broad surveys synthesize techniques and opportunities in AI in finance (Cao 2022); bibliometric and content reviews identify major research clusters across banking, markets, risk, and financial services (Goodell et al. 2021; Ahmed et al. 2022; Bahoo et al. 2024; Cao et al. 2024); and adjacent reviews emphasize the role of big

data, reinforcement learning, and large language models in widening the financial AI toolkit (Goldstein et al. 2021; Hambly et al. 2023; Li et al. 2023b; Lee et al. 2025).

This section deliberately follows the conventional method–data–scenario structure, but not in order to provide a comprehensive technical survey of financial AI. That task has been addressed by existing reviews. Here, the structure serves a different purpose: it traces how AI evolved from a set of analytical tools into a set of components that increasingly participate in human–AI hybrid financial decision systems. The method view shows how AI moved from scoring and prediction to representation learning, sequential control, language-based reasoning, and bounded agency. The data-object view shows how the evidentiary basis of financial decisions expanded from structured variables to text, networks, alternative data, and AI-generated decision artifacts. The scenario view shows how these capabilities enter different workflows, where they create distinct needs for human judgment, oversight, and accountability.

Traditional machine learning became influential when the core problem was to score, classify, and rank structured cases such as borrowers, firms, securities, and transactions. Deep learning gained momentum when financial information became sequential, textual, image-like, or high-dimensional. Reinforcement learning rose when finance researchers reframed some problems as dynamic control under feedback rather than static prediction. FinLLMs and GenAI have expanded the frontier again by turning language into an interface for reasoning, summarization, recommendation, and tool-using action (Wu et al. 2023; Yang et al. 2023; Islam et al. 2023; Xie et al. 2024; Krumdick et al. 2024; Yang et al. 2024; Li et al. 2025). The rise of AI in finance is therefore also a shift in the economic role of computation: from back-end scoring engine to front-end collaborator and, in some settings, to bounded decision agent.

By method: from scoring models to FinLLMs and agents

Traditional machine learning: scoring, classification, and ranking

The first durable wave of AI in finance grew out of statistical prediction and structured-case classification. Classical failure-prediction studies such as Beaver (1966) and Ohlson (1980) established the basic logic of turning accounting and repayment information into bankruptcy or default signals. Modern machine learning extended this tradition with support vector machines, random forests, gradient boosting, Bayesian classifiers, and related supervised learners that fit naturally with accounting ratios, repayment histories, transaction summaries, and customer attributes (Henrique et al. 2019; Lessmann et al. 2015; Nazareth and Reddy 2023; Zhou and Shen 2023). These models remain foundational because many high-volume financial decisions are still tabular, labeled, repeated, and operationally constrained.

Their attraction in finance is not only predictive. Tree ensembles and related tabular learners can often be validated, benchmarked, stress-tested, and monitored more easily than more opaque systems. In lending, fraud detection, and transaction monitoring, they convert repeated cases into scalable scores or ranked queues, while leaving humans to set thresholds, handle exceptions, judge fairness, and document adverse decisions. In human–AI hybrid finance, traditional machine learning is therefore best understood as the technology that turned borrowers, firms, transactions, and alerts into *scorable cases*.

Deep learning: representation learning for sequences, text, and complex nonlinearities

Deep learning widened the usable information set. Financial problems increasingly involved temporal dependence, nonlinear interactions, sparse signals, and raw data objects that were awkward to summarize with handcrafted variables. Recurrent and long short-term memory architectures have been used for stock return and volatility sequences (Fischer and Krauss 2018); convolutional architectures have been applied to chart-like and order-book representations (Chen and Tsai 2020); and flexible representation learning has improved empirical asset pricing when many characteristics and interactions are present (Gu et al. 2020). In finance, the economic significance of deep learning lies less in any single architecture than in its ability to learn features from sequential, textual, or multimodal inputs that exceed the expressive range of classical scorecards.

This broader representational capacity comes with a different burden of oversight. A tabular model may produce a wrong but stable ranking; a deep model may fail because the representation itself shifts, because the training distribution becomes stale, or because latent features are difficult for practitioners to inspect. For that reason, deep learning in finance raises the importance of robustness checks, regime sensitivity, challenger models, and human interpretation of whether a learned signal makes economic sense. Better representation learning often increases the need for stronger validation and review.

Reinforcement learning: from prediction to sequential control

Reinforcement learning (RL) and deep reinforcement learning (DRL) move finance AI from prediction toward sequential control. In trading, execution, hedging, market making, and portfolio rebalancing, the output is not merely a score but a policy mapping states to actions (Hambly et al. 2023). This makes RL especially important for a human–AI hybrid finance review because delegation becomes unavoidable. Once a model outputs an action policy, the organization must decide which actions are automatic, which require approval, which can be overridden, and what safety constraints must remain outside the model.

The key issue is therefore not only whether RL improves simulated or backtested performance, but whether the resulting policy can be supervised in live environments characterized by nonstationarity, market impact, and strategic interaction. In financial settings, RL systems often require hard position limits, real-time kill switches, offline evaluation, and independent monitoring. RL is thus a methodological turning point because it forces the discussion from prediction quality to decision rights and control design.

FinLLMs and generative AI: language, workflow, and AI-generated decision artifacts

FinLLMs and GenAI extend AI into language, interaction, and workflow. Systems such as BloombergGPT, FinGPT, FinanceBench, BizBench, FinBen, FinRobot, and InvestorBench signal a move from narrow text classification toward financial reasoning, grounded retrieval, report drafting, question answering (Q&A), tool use, and agentic decision support (Wu et al. 2023; Yang et al. 2023; Islam et al. 2023; Krumdick et al.

2024; Xie et al. 2024; Yang et al. 2024; Li et al. 2025). This is a qualitative shift because finance is document-heavy: analysts read filings, compare management narratives, write research notes, answer client questions, justify approvals, and document compliance. FinLLMs therefore affect not only back-end prediction but also front-end collaboration.

FinLLMs also introduce *AI-generated decision artifacts*: summaries, retrieval traces, tool-call results, draft memos, Q&A transcripts, red-flag lists, and agent logs that shape downstream human decisions. Unlike prices, filings, or transaction records, these artifacts are synthesized outputs. They are not raw data objects in the same sense as statements or news; they are intermediate evidentiary objects that shape downstream human decisions. This makes source grounding, logging, approval gating, and provenance especially important. In human–AI hybrid finance, the question is no longer only what data the model consumes, but also what decision-relevant artifacts the model produces for humans to review, trust, challenge, or act upon.

The four method families in Table 2 should be read as a cumulative sequence rather than a replacement sequence. Traditional machine learning did not disappear when deep learning arrived; many credit, fraud, and collections settings remain tabular, label-rich, and operationally constrained. Deep learning did not eliminate econometric reasoning; it expanded the representation space. RL did not replace forecasting; it reframed some financial problems as sequential decision problems under constraints. FinLLMs did not merely add a new text classifier; they changed the interface through which humans search, read, justify, and act. For this reason, a method taxonomy in finance should always be linked to decision rights. Different methods do not just imply different accuracy profiles; they imply different possibilities for intervention, override, accountability, and workflow design.

By data object: from tabular variables to multimodal signals

If method families explain how AI spread in finance, data objects explain why. Structured numerical data remain the backbone of financial AI: accounting variables, repayment histories, transaction streams, balance-sheet items, order-flow records, and

Table 2 Method families in finance and their human–AI hybrid implications

Method family	Typical financial task	Strength	Implication for human–AI hybrid finance
Traditional machine learning	Credit scoring, default screening, fraud triage, case ranking	Stable performance on structured and repeated cases	Human review concentrates on thresholds, exceptions, fairness, and adverse-action logic
Deep learning	Forecasting, asset pricing, multimodal classification, text and sequence analysis	Learns representations from complex, nonlinear, or high-dimensional data	Expands the information set but raises validation, robustness, and interpretation burdens
RL/DRL	Execution, hedging, market making, portfolio rebalancing	Learns policies for sequential decisions under feedback	Shifts the key question from prediction to delegation, constraint design, and interruptibility
FinLLMs/GenAI	Summarization, Q&A, research drafting, retrieval, tool use, bounded agents	Turns language into both data and interface; creates AI-generated decision artifacts	Requires source grounding, logging, approval rules, and verification of AI-generated decision artifacts

customer attributes map directly onto institutional routines in underwriting, pricing, collections, fraud screening, and stress testing. These data are comparatively stable, auditable, and economically interpretable. This is one reason why much of financial AI remains anchored in tabular modeling even as newer methods proliferate.

The rise of textual data transformed the information environment. News, filings, earnings calls, analyst reports, complaints, and social media contain information about sentiment, tone, uncertainty, disclosure quality, litigation risk, and managerial expectations. Early work showed that language contains market-relevant information (Tetlock 2007), and later surveys established textual analysis as a standard component of finance research (Loughran and McDonald 2020). Transformer-based systems and FinLLMs have extended this line from dictionaries and bag-of-words to retrieval, summarization, Q&A, and cross-document comparison (Li et al. 2023b; Lee et al. 2025).

Network and relational data matter because many financial objects are not independent. Small and medium-sized enterprise (SME) credit, payment fraud, anti-money laundering (AML), supply-chain risk, interbank exposure, and systemic fragility depend on connections rather than isolated attributes. Graph-based approaches are therefore increasingly important in settings where contagion, affiliation, hidden linkage, or relational opacity matters (Goldstein et al. 2021; Lu et al. 2024). Multimodal and alternative data widen the frontier further. Satellite images, web traffic, geolocation, job postings, app behavior, environmental, social, and governance (ESG) disclosures, and other digital traces may reveal economic activity before it appears in statements or prices (Goldstein et al. 2021; Sun et al. 2024; Cao et al. 2024). These inputs are especially valuable in thin-file or rapidly changing settings, but they also raise harder questions about privacy, provenance, proxy bias, and stability.

Table 3 also highlights a deeper issue: expanding the data frontier expands the *interpretive burden*. When inputs are accounting ratios or repayment histories, financial professionals usually understand the semantics of each variable. When inputs are embeddings, graph centralities, satellite-derived proxies, telematics, or multimodal representations, expertise shifts upstream and downstream. Upstream, professionals help define which data sources are legitimate, stable, and permissible. Downstream, they act

Table 3 Data objects in finance and the main human verification tasks they create

Data object	What it adds	Main risk	Human verification focus
Structured numerical data	Comparable, repeated, and audit-friendly signals for scoring and monitoring	Proxy bias, omitted interactions, brittle thresholds, drift	Threshold review, feature plausibility, fairness checks, and recourse logic
Textual data	Narrative disclosure, beliefs, tone, soft information, and document-intensive context	Ambiguity, stale context, source noise, persuasive but weak narratives	Source grounding, temporal alignment, factual consistency, and interpretation of economic relevance
Network and relational data	Spillovers, hidden ties, contagion channels, and cross-entity dependence	Incomplete coverage, unstable edge definitions, overreading relational structure	Coverage checks, causal plausibility, cross-entity validation, and escalation rules
Multimodal and alternative data	Early or otherwise unavailable signals beyond statements and prices	Representation drift, privacy risk, provenance uncertainty, proxy discrimination	Provenance, consent, robustness, legitimacy of use, and challenge rights

as *epistemic arbitrators*: they decide whether a signal is merely predictive or is also suitable as decision-relevant evidence.

This distinction between predictive signal and decision-relevant evidence is crucial in finance. A variable may improve default prediction but still be inappropriate for credit decisions if it proxies for protected characteristics, violates privacy expectations, or cannot be explained to an applicant. A telematics signal may help an insurer price risk but still require scrutiny if it proxies for location, occupation, or other sensitive patterns. A social-media signal may help forecast market attention but be unstable during crises or vulnerable to manipulation. A network feature may help detect suspicious behavior while also raising guilt-by-association concerns. The spread of AI in finance therefore widens not only the information set but also the governance problem: provenance, consent, temporal alignment, fairness, and contestability become part of model quality.

By scenario: banking, markets, asset management, and oversight

Across scenarios, the most mature AI applications remain in credit allocation and risk management. Bankruptcy prediction, credit scoring, underwriting, fraud detection, and early warning are natural use cases because labels exist and economic value can be quantified (Henrique et al. 2019; Lessmann et al. 2015; Nazareth and Reddy 2023). A second major cluster concerns asset pricing, return forecasting, and portfolio allocation, where machine learning helps process high-dimensional predictors and nonlinear dynamics (Gu et al. 2020). A third cluster concerns market microstructure and trading, including order-book prediction, execution, and market making, although live deployment remains constrained by nonstationarity, competition, and endogenous market impact (Fischer and Krauss 2018; Hambly et al. 2023; Budish et al. 2015; Menkveld 2013). A fourth cluster concerns text-intensive intermediation—research, advisory, accounting, investor relations, compliance, and regulatory technology (RegTech)—where AI increasingly supports retrieval, summarization, recommendation, and documentation (Cao et al. 2024; Li et al. 2023b; Lee et al. 2025). A fifth cluster concerns insurance, RegTech, supervisory technology (SupTech), and systemic oversight, where AI helps monitor exposures, summarize filings, and prioritize supervisory attention (Petrone et al. 2022; Cao 2022) (Table 4).

The scenario view matters because human–AI hybrid finance problems are domain-specific. Banking is where individual-level consequences are most visible: borrowers may experience AI-assisted decisions as denials, repricing, documentation requests, or explanations. Research on AI in banking should therefore connect model metrics to borrower outcomes, employee behavior, and institutional governance rather than stop at predictive lift. Capital markets present a different problem. Here AI is not only a forecasting technology but an equilibrium technology: widespread use of similar signals may change prices, liquidity, crowding, and the very environment on which models rely (Kleinberg and Raghavan 2021). Robo-advisory and investment guidance sit somewhere between automation and fiduciary communication: the challenge is not only scaling advice, but ensuring suitability, understandable communication, and long-run investor welfare (Cardillo and Chiappini 2024; Greig et al. 2025). Fraud and AML systems, by contrast, are often triage systems whose value depends on whether they improve the ordering of scarce human attention.

Table 4 Major financial scenario families and their human–AI hybrid governance challenges

Scenario family	Representative tasks	Outcome observability	Main human–AI hybrid challenge
Retail and SME lending	Screening, default prediction, pricing, document review, collections triage	Delayed and partly censored	Combine model output with soft information, fairness constraints, recourse, and relationship context
Payments, fraud, and AML	Alert generation, anomaly detection, investigation routing, suspicious-case escalation	Often depends on investigation quality	Improve attention allocation without overwhelming reviewers or creating hidden false negatives
Capital markets and portfolio management	Forecasting, factor discovery, portfolio construction, surveillance	Market outcomes are fast but endogenous	Separate genuine signal from equilibrium feedback, crowding, and strategic adaptation
Trading and execution	Routing, execution timing, market making, hedging, limit monitoring	Immediate but hard to attribute counterfactually	Constrain automation, maintain interruptibility, and supervise policies under stress
Research, advisory, and investment banking	Retrieval, disclosure analysis, memo drafting, client communication, valuation support	Long-horizon and partly qualitative	Ensure grounding, suitability, and retention of human judgment in communication-heavy decisions
Insurance, RegTech, SupTech, and systemic oversight	Claims triage, underwriting support, filing review, institution monitoring, intervention support	Mixed and often organization-level	Balance auditability, fairness, legitimacy, and spillover risk across institutions and stakeholders

Insurance provides another useful contrast. An AI system may estimate claim severity from images, flag potential claims fraud, recommend underwriting classes, or support usage-based pricing from telematics and other behavioral data. Yet an adjuster or underwriter often must interpret policy language, evaluate disputed evidence, communicate reasons to policyholders, and handle contestation. Insurance therefore highlights why predictive lift must be combined with decision-useful explanation, fairness review, and meaningful oversight.

RegTech and SupTech extend the human–AI hybrid problem from private financial institutions to compliance and public oversight. Within firms, AI may summarize regulatory changes, map obligations to business processes, draft compliance memos, or prioritize surveillance cases. For supervisors, AI may screen regulatory filings, identify anomalous liquidity or capital patterns, and rank institutions or markets for review. These systems rarely make final regulatory judgments on their own, but they shape attention, escalation, and the allocation of supervisory resources.

A final cross-scenario distinction concerns the observability of outcomes. Credit outcomes are delayed and partly censored because rejected applicants do not reveal repayment behavior. Trading outcomes arrive quickly but are endogenous and counterfactual: one rarely observes what an unexecuted trade would have earned. Fraud, AML, and insurance claims outcomes often depend on whether a flagged case is investigated thoroughly. Advisory outcomes may matter only over long horizons through realized investor welfare. Supervisory outcomes are even harder to evaluate because successful oversight often prevents events that are not directly observed. This is why “AI application” cannot be separated from workflow design. Outcome observability shapes how quickly humans learn from AI, how overrides are evaluated, and how organizations decide whether humans and AI are genuinely complementary.

The rise of AI changed finance along three linked fronts: method frontier, data frontier, and application frontier. In this review, these frontiers are not treated as endpoints of a technical survey. They are treated as the conditions that make a decision-system perspective necessary. Once AI produces recommendations, actions, summaries, rankings, alerts, or AI-generated decision artifacts, the central question is no longer only what the model can predict or optimize. The central question becomes how humans and AI should jointly decide, who has authority at each stage, how reliance should be calibrated, how explanations should support accountability, and how AI-mediated actions reshape future decisions.

AI’s challenge: when human–AI collaboration works, and when it fails

The most tempting assumption in human–AI hybrid finance is that human judgment plus AI prediction should outperform either humans or AI alone. This assumption is plausible because the two sides appear to have complementary strengths: AI brings scale, consistency, computational power, and pattern recognition, while humans bring contextual knowledge, strategic reasoning, moral responsibility, and institution-specific information. Yet the evidence on human–AI collaboration gives a more cautious lesson. Human–AI hybrid financial decision systems create value only when the task, model, interface, human role, incentives, and feedback process are jointly designed. The

central challenge is therefore not merely model accuracy; it is the allocation of attention, authority, and accountability.

A useful distinction is between *model performance* and *hybrid decision performance*. Model performance asks whether an AI system predicts, classifies, ranks, or summarizes well in a benchmark. Hybrid decision performance asks whether a human–AI workflow improves economic outcomes after accounting for reliance, override, cost, time pressure, accountability, and downstream effects. A model can improve predictive accuracy while worsening decisions if users misunderstand its scope, overweight its recommendations, ignore private information, or place it at the wrong stage of the workflow. Conversely, a model with modest accuracy may improve decisions if it helps route scarce human attention to the cases where human judgment is most valuable.

This section develops a boundary-condition view of human–AI collaboration in finance. It integrates the broader human–AI literature with finance-specific mechanisms and advances the *human–AI hybrid finance puzzle*: if humans and AI have different strengths, why do their combinations so often fail to outperform the best stand-alone human or AI decision maker? The answer is that complementarity is not a property of humans and AI in isolation. It is a property of a decision architecture.

This section focuses on the behavioral and organizational conditions under which human–AI combinations improve or degrade financial decisions. It treats delegation, handoff, reliance, incentives, and feedback as conditions for complementarity rather than as governance mechanisms in themselves. The governance and explanation mechanisms that respond to these conditions are developed in “[Making human–AI hybrid finance work: explainability, trust, and governance](#)” section, while their future workflow and market-level implications are examined in “[The future of human–AI hybrid finance](#)” section.

Empirical evidence and the human–AI hybrid finance puzzle

The broader evidence base is now large enough to discipline optimistic claims about synergy. A recent systematic review and meta-analysis finds that human–AI combinations are often useful, but they do not automatically outperform the better of humans or AI alone; losses are particularly common in decision tasks, whereas gains are more visible in creation tasks (Vaccaro et al. 2024). This result is directly relevant to finance. Financial AI systems are rarely deployed merely to create text or produce exploratory outputs. They are used to screen borrowers, rank alerts, recommend trades, rebalance portfolios, summarize disclosures, flag misconduct, justify decisions, or support insurance and supervisory review. These are consequential decision tasks in which errors can be asymmetric, delayed, legally salient, or amplified through markets.

The human–AI hybrid finance puzzle should not be read as a claim that human–AI collaboration is ineffective. Rather, it asks when collaboration is truly complementary. Studies of productive delegation show that performance depends on who is allowed to defer to whom and at what point in the sequence (Fügener et al. 2022). Learning-to-defer research formalizes this idea by training models to decide when to act and when to route cases to humans (Madras et al. 2018; Mozannar and Sontag 2020; Wei et al. 2024; Montreuil 2026). For finance, the practical analogue is deciding which credit files should be automatically approved, which should be escalated to underwriters, which AML

alerts require human investigation, which insurance claims require adjuster review, and which trading or portfolio recommendations should be executable rather than advisory.

Human and AI errors also differ in ways that matter for financial decisions. AI usually excels at scale, consistency, and pattern extraction from large data volumes. Humans may retain advantages in novelty, exceptions, strategic behavior, moral ambiguity, and local institutional knowledge (Gonzalez and Heidari 2025; Natarajan et al. 2025). In credit, a model may outperform loan officers on average, while officers may still add value for thin-file borrowers, temporary cash-flow shocks, or exceptional collateral. In insurance, an image-based claims model may estimate damage severity accurately, while a claims adjuster remains important for disputed coverage, policy interpretation, and claimant communication. In capital markets, AI may process news and filings faster than analysts, while human experts may better interpret regime shifts, policy events, or strategic disclosure.

The form of assistance also matters. Uncertainty displays, prediction sets, and human-aligned calibration can support better reliance when users understand what the uncertainty means and have authority to act on it (Corvelo Benz and Rodriguez 2023; Cresswell et al. 2024; Marusich et al. 2024). Explanations, by contrast, are not automatically beneficial. They can improve understanding and contestability, but they can also create unwarranted confidence if they are unstable, decorative, or detached from the model's true reasons (Bansal et al. 2021). The empirical lesson is therefore not “add AI” or “add explanations.” It is to design the assistance paradigm around the financial task.

Decision paradigms, the delegation frontier, and handoff failures

We define the *delegation frontier* as the stage-specific boundary at which decision authority shifts between humans and AI within a financial workflow. Human–AI hybrid decision performance depends heavily on the role AI plays in the decision process. In finance, AI assistance is not a single intervention. It can operate as a screening aid, a second opinion, a ranked recommender, a conversational copilot, or an autonomous executor. These paradigms differ in who acts first, who retains discretion, what information is displayed, and where accountability resides. Treating them as interchangeable would obscure the core authority-allocation problem (Table 5).

This taxonomy clarifies why evidence from one setting may not transfer to another. A model that is useful as a screening aid may be dangerous as an autonomous executor. A conversational copilot that improves analyst productivity in summarizing filings may not be reliable enough to generate unverified investment recommendations. A ranked recommender may improve attention allocation in fraud detection, yet create anchoring if investigators treat the rank as a substitute for evidence. Thus, the relevant design question is not simply whether AI should augment or automate. It is where the *delegation frontier* should be drawn across information acquisition, recommendation, approval, execution, and monitoring.

For example, in a trading execution workflow, an AI system may recommend the timing, venue, or order-splitting strategy for a trade, while an execution algorithm routes child orders automatically within pre-set risk limits. The trader may still retain authority over the trading objective, limit changes, and kill-switch activation. The delegation

Table 5 AI assistance paradigms in human–AI hybrid financial decisions

AI role	Human role	Typical financial uses	Main failure mode
Screening aid	Allocate attention and inspect exceptions	AML alerts, fraud triage, credit screening, cyber-risk monitoring	Missed exceptions if thresholds or labels are poorly calibrated
Second opinion	Challenge, confirm, or question an initial human judgment	Investment research, credit committee review, model risk review	Disagreement ignored or treated as a procedural formality
Ranked recommender	Prioritize alternatives and structure choice sets	Securities selection, customer leads, suspicious transactions, borrowers	Anchoring on ranks; neglect of tail risk or unmodeled context
Conversational copilot	Search, summarize, draft, explain, and document	Filing analysis, client communication, compliance documentation, research memos	Fluent hallucination, over-trust, and substitution for original analysis
Autonomous executor	Act within bounded rules or policies	Trade execution, liquidity provision, rebalancing, monitoring workflows	Accountability gaps, correlated actions, drift, and insufficient override

frontier is therefore not a binary choice between human and AI, but a stage-specific boundary between recommendation, bounded execution, and human intervention.

A similar logic applies to AML alert escalation. An AI system may rank alerts, group related transactions, and draft an initial case narrative; an analyst may investigate supporting evidence and decide whether to escalate; and a compliance officer may retain final authority over whether to file a suspicious activity or transaction report. Here, the delegation frontier runs across screening, investigation, escalation, and final reporting rather than across a single yes-or-no decision.

This stage-specific view is especially important for FinLLMs and agentic systems. Benchmarks such as FinBen, BizBench, and InvestorBench show rapid progress in financial language understanding, reasoning, and decision-oriented evaluation (Xie et al. 2024; Krumbick et al. 2024; Li et al. 2025). However, a system that performs well on question answering or document analysis may not be suitable for bounded execution. The same model can be low-risk as a retrieval assistant, moderate-risk as a recommendation agent, and high-risk as an executor connected to trading, lending, compliance, or claims tools. Human–AI hybrid finance therefore requires action-centered evaluation, not just text-centered evaluation.

Even when humans and AI possess complementary capabilities, coordination is difficult. Several dilemmas recur. First, users often receive recommendations without a clear mental model of when the system is reliable. Second, developers often optimize average model performance, whereas financial organizations care about tail losses, reputational exposure, legal defensibility, and auditability. Third, local rationality can create system-level risk: a loan officer may overrule a model to protect fairness in one case, while a risk committee may prefer consistency across thousands of cases. Fourth, workflows may be nominally human-in-the-loop while functioning in practice as AI-led systems with the human reduced to a procedural approver rather than an active decision maker (Lee and See 2004; Natarajan et al. 2025).

These dilemmas imply that many financial AI failures are not prediction failures; they are *handoff failures*. A model may be accurate on average but be routed to the wrong

user, surfaced at the wrong time, framed with the wrong confidence cue, or embedded in a workflow that discourages thoughtful review. In a lending workflow where underwriters are evaluated on throughput, even a well-calibrated score can generate poor decisions if officers have no time to inspect borderline cases. In a trading workflow where AI outputs are delivered as point forecasts without scenarios, users may over-anchor on a single number. In compliance workflows with thousands of alerts, explanation quality, prioritization, and escalation paths may matter as much as predictive power.

For instance, an AML model may correctly identify a high-risk cluster of transactions, but the value of that signal can be lost if the alert is routed to an overloaded reviewer without source provenance, case history, or a clear escalation path. Similarly, a trading model may produce a useful execution signal, but hybrid decision performance can deteriorate if the signal is displayed as a point recommendation without liquidity conditions, uncertainty, or stress-state warnings.

Finance adds a further complication: feedback is often delayed, censored, or endogenous. Rejected loan applicants do not reveal repayment outcomes. Unexecuted trades do not reveal counterfactual profits. Dismissed claims or AML alerts may not reveal hidden misconduct unless later events trigger further investigation. A portfolio manager may never learn whether an ignored AI recommendation would have improved risk-adjusted returns. These missing counterfactuals make it difficult for humans to learn when AI is reliable and for organizations to estimate the value of human override. They also create selection problems for researchers: observed outcomes are partly the product of prior human–AI decisions.

For this reason, human–AI hybrid finance should be studied as a workflow-design and learning problem. Disagreement between humans and AI should not be treated only as noise. It can reveal private human information, model blind spots, incentive distortions, or cases outside the training distribution. If disagreement is logged and outcomes are later observed, institutions can estimate when overrides add value. If disagreement is suppressed, institutions lose one of their best sources of learning about model limitations.

Reliance failures and the reliance wedge

We define the *reliance wedge* as the gap between actual reliance on AI and warranted reliance given relative accuracy, uncertainty, incentives, and accountability. Reliance is the behavioral core of human–AI hybrid finance. The automation literature distinguishes use, misuse, disuse, and abuse of automated systems, and emphasizes that appropriate reliance is a design problem rather than a personality trait (Parasuraman and Riley 1997; Parasuraman et al. 2000; Lee and See 2004). In financial settings, both too much and too little reliance can destroy value. *Overreliance* occurs when users accept AI advice beyond what its reliability justifies. Analysts may anchor on AI-generated summaries; credit officers may interpret model scores as objective facts; portfolio managers may mistake backtest stability for structural validity. *Underreliance* occurs when users reject useful AI because of opacity, salient errors, status concerns, or poor interface design.

A credit-underwriting example illustrates the wedge. Overreliance may occur when a loan officer treats a high AI credit score as sufficient evidence for approval and fails to inspect recent income disruption, unusual collateral, or possible data-quality problems.

Underreliance may occur when the officer discounts a model warning because the applicant is a long-standing relationship customer, even though the model identifies repayment patterns that the officer has not observed. In both cases, the issue is not whether the credit score is useful in general, but whether the officer's reliance is warranted for this borrower, this context, and this decision.

Classic studies of algorithm aversion show that people may reject algorithms after observing mistakes, even when the algorithm outperforms human judgment on average (Dietvorst et al. 2015). Later work documents algorithm appreciation in some settings, indicating that people may prefer algorithmic advice when it appears objective or precise (Logg et al. 2019). Finance contains both tendencies. A professional may follow AI to shift blame, because “the model recommended it.” Another may override AI to preserve discretion or demonstrate expertise. A manager may prefer AI because it standardizes decisions and reduces cost, while a regulator may worry that standardization hides correlated risk.

The key objective is therefore not global trust but *selective reliance*. Global reliance asks whether users trust the system overall. Selective reliance asks whether users rely on AI when it is likely to be right and withhold reliance when it is likely to be wrong. This distinction is central to the reliance wedge. A trader should not rely on a model equally in liquid and illiquid markets. A credit officer should not rely on a score equally for thick-file and thin-file borrowers. An insurance underwriter should not rely on a risk classification equally when the input data are complete and when telematics coverage is sparse. An analyst should not rely on an LLM equally for factual retrieval, causal inference, and valuation judgment.

Design interventions can narrow the reliance wedge, but they can also introduce new costs. Cognitive forcing can reduce overreliance by requiring users to form an initial judgment before viewing AI advice, yet it may slow workflows or create fatigue if applied indiscriminately (Bucinka et al. 2021). Confidence displays, explanations, source grounding, and case-similarity indicators can support selective reliance, but only when users understand them and have authority to respond. In high-volume finance workflows, the challenge is to calibrate reliance without turning oversight into friction or rubber-stamping.

Boundary conditions and research questions

The literature can be synthesized into a set of boundary conditions. Human–AI complementarity is more likely when human and AI errors are not highly correlated, humans possess information the model lacks, uncertainty is communicated in a decision-useful form, the human has authority and incentive to override, and the task allows enough time for meaningful review. It is less likely when AI already dominates human performance, humans cannot detect model errors, explanations create false confidence, incentives encourage blame-shifting, or time pressure turns oversight into a rubber stamp (Table 6).

This boundary-condition view suggests that the phrase “human in the loop” is too vague for finance. A human can be an informed reviewer, a symbolic approver, a private-information provider, a moral and regulatory check, a liability holder, or an overloaded monitor. These roles have different implications for performance and

Table 6 Boundary conditions for human–AI complementarity in finance

Boundary condition	Hybrid decisions are more likely to work when...	Hybrid decisions are likely to fail when...
Task structure	Tasks are repeated, outcomes are measurable, and losses can be linked to decisions	Tasks are novel, strategic, rare, ambiguous, or subject to regime shifts
Information complementarity	Humans possess private, local, or contextual information absent from model inputs	Human judgment duplicates model inputs or is dominated by bias and noise
Authority allocation	Decision rights match comparative advantage; overrides are logged and reviewed	Humans are nominally in the loop but lack authority, time, expertise, or accountability
Feedback quality	Outcomes are timely, observable, and connected to human and AI choices	Feedback is delayed, censored, endogenous, or contaminated by selection
Explanation and uncertainty	Explanations, confidence, and domain-coverage cues support selective reliance	Explanations are decorative, manipulative, unstable, or create false reassurance
Incentives and accountability	Users are rewarded for decision quality and responsible override	Users follow AI to avoid blame or reject AI to preserve autonomy

Table 7 Research questions for human–AI collaboration in finance

Problem	Research question	Example financial setting
Complementarity	When do professionals add information unavailable to the model, and when do they add only noise?	Thin-file credit, relationship lending, regime-shift detection
Delegation frontier	When should AI recommend, when should it execute, and when should it defer to humans?	Trading execution, credit approval, compliance escalation
Override value	When should human override be encouraged, discouraged, or audited?	Credit committees, model risk management, portfolio rebalancing
Selective reliance	Which uncertainty, explanation, and interface designs reduce the reliance wedge?	Robo-advice, AML alerts, investment research, risk dashboards
Learning and feedback	Which workflows transform disagreement into learning rather than suppressing it?	Audit logs, shadow deployment, model validation, post-decision reviews
System effects	When do locally effective human–AI decisions generate correlated behavior or systemic risk?	Algorithmic trading, credit allocation, asset management, liquidity provision

governance. Human–AI hybrid finance should therefore be treated as a mechanism-design problem: institutions design thresholds, review queues, escalation paths, override rules, explanation formats, feedback loops, and accountability assignments, while researchers test which designs reduce the reliance wedge and improve welfare (Table 7).

These questions shift the focus from whether humans or AI are better to which allocation of information, authority, incentives, and feedback creates the best human–AI hybrid financial decision system. “AI’s challenge: when human–AI collaboration works, and when it fails” section has therefore treated reliance, delegation, override, and handoff as behavioral and organizational conditions for complementarity. “Making human–AI hybrid finance work: explainability, trust, and governance” section turns from these conditions to the mechanisms that can make hybrid financial decisions explainable, governable, and accountable. If human–AI hybrid finance depends

on selective reliance, authority allocation, and meaningful review, then explainability, trust calibration, and governance are not optional add-ons. They are core design requirements.

Making human–AI hybrid finance work: explainability, trust, and governance

The previous section identified the behavioral and organizational conditions under which human–AI collaboration succeeds or fails. This section turns to the mechanisms that make such collaboration governable in high-stakes financial settings. Explainability, trust calibration, and governance are not add-ons to human–AI hybrid finance. They are control mechanisms that help determine whether AI-mediated decisions can be understood, contested, audited, and assigned to accountable parties. A black-box credit model may deny or reprice a loan; a trading model may create losses; an AML model may escalate a customer; an LLM may generate a misleading investment memo; and an agent may take actions that are difficult to reconstruct.

Black boxes, justification gaps, and high-stakes financial decisions

Finance is unusually sensitive to black-box AI because many financial decisions allocate scarce opportunities, impose costs, and create legal or prudential obligations. A model output often needs to be translated into an institutional reason: why a loan was denied, why a client was escalated, why a claim was rejected, why a trade was executed, why a portfolio was rebalanced, or why a compliance case was closed. Surveys of XAI in finance therefore emphasize that explainability is a functional requirement for adoption, contestability, and model risk management rather than a cosmetic interface feature (Weber et al. 2024; Černevičienė and Kabašinskas 2024; Mersha et al. 2024; Arrieta et al. 2020).

The black-box problem is not simply about nonlinear models. It arises whenever there is a *justification gap*: a gap between the logic by which an AI system produces an output and the logic by which a human user or financial institution can defend an action. A highly accurate deep model may still fail organizationally if a credit committee cannot explain it, a relationship manager cannot communicate it, an auditor cannot reproduce it, or a regulator cannot evaluate it. Conversely, a less complex model may be preferable when it supports contestability, auditability, and legal defensibility in a high-stakes setting. The central question is therefore not whether all financial AI should be simple. It is which parts of the decision system require transparency, to whom, for what purpose, and with what level of detail.

This distinction matters because financial users are heterogeneous. A customer may need a comprehensible reason and feasible recourse; a front-line employee may need a concise case-level explanation and a warning when a case falls outside the model's reliable domain; a model risk team may need stability tests, challenger-model comparisons, drift reports, and fairness diagnostics; a senior manager may need aggregate risk, incident, and performance dashboards; and a regulator may need documentation showing that the institution has governance processes rather than merely an explanation interface. Treating all these needs as “interpretability” collapses distinct functions that should be designed and evaluated separately.

From XAI methods to decision-useful explanations

We define *decision-useful XAI* as explanation that improves actionability, auditability, contestability, and accountability in financial decisions. XAI research offers several tool families. Inherently interpretable models use sparse rules, monotonic constraints, scorecards, or transparent trees, and are especially valuable when legal defensibility and stable reasoning are central (Rudin 2019). Post-hoc local methods such as Local Interpretable Model-agnostic Explanations (LIME) and Shapley Additive Explanations (SHAP) approximate or decompose individual predictions, helping users identify which variables shaped a particular recommendation (Ribeiro et al. 2016; Lundberg and Lee 2017). Counterfactual explanations and recourse methods ask what changes would alter an outcome, making them particularly relevant for credit, insurance, and consumer finance (Wachter et al. 2018; Ustun et al. 2019). Rule extraction, distillation, example-based explanations, saliency methods, and natural-language rationales provide additional routes, but their value depends on the decision context and user role (Arrieta et al. 2020; Byrne 2023).

For human–AI hybrid finance, however, the key criterion is not whether an explanation is technically elegant. It is whether the explanation is *decision-useful*. A decision-useful explanation should satisfy four tests. First, it should be *faithful enough* for the decision purpose: it should not provide a plausible story unrelated to the model's behavior (Jacovi and Goldberg 2020). Second, it should be *understandable* to the relevant user, which may require different formats for customers, analysts, validators, and regulators (Liao et al. 2020). Third, it should be *actionable* when the decision invites contestation or recourse (Ustun et al. 2019). Fourth, it should be *auditable*: the explanation and the human response to it should be recorded in a way that supports later review (Cobbe et al. 2021).

This view changes how XAI should be evaluated. A feature-attribution chart may be useful for a model validator but insufficient for a rejected borrower. A counterfactual explanation may be useful for a customer but inadequate for a model risk committee evaluating distribution shift. A portfolio manager may need scenario sensitivity, factor exposures, and narrative drivers rather than a generic variable-importance plot. A compliance officer may need case similarity, alert history, and source provenance. Therefore, XAI benchmarks for finance should measure not only explanation fidelity or user satisfaction, but also whether explanations improve decision quality, reduce inappropriate reliance, support recourse, and strengthen auditability.

Insurance illustrates this distinction. If an AI system recommends denying a claim or assigning a policyholder to a higher-risk underwriting class, a feature-attribution chart may be insufficient for the affected customer. A decision-useful explanation should connect the recommendation to policy terms, relevant evidence, uncertainty, and available challenge or recourse channels. At the same time, the explanation may need to avoid revealing the full logic of fraud-detection rules, because excessive disclosure could enable strategic manipulation. Insurance therefore shows why financial XAI must balance contestability, customer communication, and system integrity.

More explanation is not always better. Explanations can increase understanding, but they can also create false reassurance, anchor users on irrelevant factors, or be manipulated to encourage acceptance. Empirical work on AI explanations shows that

explanations do not automatically improve complementary team performance and may sometimes increase overreliance (Bansal et al. 2021; Schemmer et al. 2022). Adversarial and behavioral studies further show that explanation interfaces can be fooled, manipulated, or optimized to shape user behavior rather than reveal model reasoning (Slack et al. 2020; Lakkaraju and Bastani 2020; Li and Yin 2024). For finance, this means explanation design must be governed, not merely deployed.

LLMs make this issue sharper because they can generate fluent rationales that sound authoritative without being tied to the actual decision process. A traditional XAI method may be limited, unstable, or incomplete, but it is at least designed to connect an output to model behavior. An LLM-generated narrative may instead summarize retrieved sources, infer a plausible reason, or produce a polished compliance paragraph that is not causally linked to the model, the data, or the approval path. This distinction between explanation and rationalization is crucial. If AI-generated rationales are stored as compliance documentation, communicated to customers, or used in investment memos, institutions need mechanisms that connect generated text to sources, model outputs, decision records, and human approvals.

Explanation governance also involves a security and strategic-behavior trade-off. Full transparency can support customer rights and regulatory oversight, but it can also help credit applicants, traders, fraudsters, or money launderers game models. The goal is therefore not maximum disclosure in every setting. It is *appropriate disclosure*: enough information to support rights, oversight, recourse, and accountability without destroying system integrity. This balance is especially delicate in finance because many actors are strategic and financially motivated. Robust governance must therefore consider fairness, privacy, security, and contestability together rather than treating them as separate ethics topics.

Trust calibration, governance stack, and meaningful oversight

Trust calibration is the behavioral objective of explanation and interface design. The aim is not high trust, but warranted trust: users should rely on AI when it is likely to improve the decision and withhold reliance when it is not. The automation literature has long distinguished appropriate use from misuse, disuse, and abuse, and trust research shows that user reliance depends on perceived reliability, transparency, workload, expertise, and institutional context (Parasuraman and Riley 1997; Parasuraman et al. 2000; Lee and See 2004; Jian et al. 2000; Bach et al. 2022; Afroogh et al. 2024). In finance, trust calibration is especially important because errors are often costly, delayed, and difficult to reverse.

This creates a layered reliance problem. A loan officer may rely on a case-level score; a risk manager may rely on aggregated model dashboards; a senior executive may rely on AI-generated summaries of model performance; and a board may rely on internal assurance reports. Each layer transforms technical evidence into a managerial representation. If any layer over-simplifies uncertainty, suppresses exceptions, or overstates model reliability, trust can become miscalibrated even if the underlying model is statistically sound.

Several mechanisms can support calibrated reliance. Confidence intervals, conformal prediction sets, uncertainty visualizations, case-similarity indicators, model cards, drift warnings, baseline comparisons, and override logs can make reliability more visible

(Cresswell et al. 2024; Corvelo Benz and Rodriguez 2023; Marusich et al. 2024). Cognitive forcing functions can slow users down when AI confidence is low or when a case is outside the model's domain (Bucinka et al. 2021). Interface experiments and behavioral models can help predict when AI advice will nudge users toward acceptance or resistance (Li et al. 2024). Yet information alone is insufficient. The display must match the user's expertise, time pressure, authority, and incentives. Building on the reliance wedge identified in "AI's challenge: when human–AI collaboration works, and when it fails" section, these mechanisms should be evaluated by whether they improve selective reliance rather than merely increasing user trust.

We define *meaningful oversight* as human supervision that has sufficient information, authority, time, independence, and responsibility to affect outcomes. Financial institutions already have traditions of model risk management, validation, documentation, stress testing, escalation, and supervisory review. Supervisory Guidance on Model Risk Management (SR 11-7) remains an important reference point for model risk management in banking (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency 2011). AI governance frameworks add concerns over transparency, fairness, privacy, accountability, robustness, human oversight, and security (National Institute of Standards and Technology 2023, 2024). Regulatory developments such as General Data Protection Regulation (GDPR) restrictions on certain automated decisions and the European Union (EU) AI Act's risk-based approach reinforce the need for explainability, contestability, and oversight in high-impact systems (European Union 2016, 2024). For finance, these concerns should be integrated with existing compliance, audit, and risk functions rather than treated as a separate ethics checklist.

A useful governance stack has three operational conditions. First, *data trustworthiness*: data must be timely, representative, well-defined, privacy-preserving, and monitored for drift, annotation noise, and multimodal conflict (Geburu et al. 2021). Second, *model reliability*: models must remain calibrated, robust, fair, and secure under distribution shift, strategic adaptation, and institutional scrutiny (Li et al. 2023a). Third, *decision feasibility*: model outputs must be translatable into lawful, explainable, workflow-compatible, value-consistent actions. A statistically sound recommendation can still fail if it cannot be defended to regulators, embedded in a workflow, accepted by users, reconciled with fairness constraints, or aligned with the institution's risk appetite (Selbst et al. 2019) (Table 8).

Meaningful oversight is easiest to state and hardest to implement. A human reviewer who sees hundreds of alerts per day may not have time for independent judgment. A committee that approves a model without understanding its data and limitations may not provide real accountability. A customer-service representative who reads an AI-generated explanation without authority to change the outcome may not provide contestability. Oversight becomes meaningful only when the human has relevant information, actual authority, enough time, independence from pressure to rubber-stamp, and responsibility for the decision. Human-in-the-loop governance should therefore be evaluated empirically through workflow data, override records, escalation outcomes, and audit trails rather than assumed from organizational charts.

The same logic applies to systemic risk supervision. A supervisory AI dashboard may rank banks, insurers, or markets by liquidity stress, capital vulnerability,

Table 8 From explanation tools to governance mechanisms

Layer	Core question	Illustrative mechanisms
Model and data transparency	What does the system use, and how stable and trustworthy are the inputs?	Data provenance, interpretable models, feature attribution, robustness tests, drift monitoring, documentation
Decision explanation	Why was this case recommended, escalated, rejected, approved, or executed?	Local explanations, counterfactuals, case similarity, uncertainty displays, source-grounded rationales
Human reliance	Did the human use AI appropriately?	Cognitive forcing, confidence calibration, training, interface audits, override logging
Accountability	Who had decision rights and responsibility?	Approval rules, audit trails, escalation paths, model-owner assignment, human approval records
Institutional governance	Is the system safe, compliant, and feasible over time?	Model validation, stress testing, privacy review, fairness monitoring, incident reporting, retirement rules
Market and system governance	Could adoption create correlated behavior or systemic vulnerabilities?	Scenario analysis, concentration monitoring, third-party risk management, supervisory reporting

interconnectedness, or unusual reporting patterns. Oversight is meaningful only if supervisors can inspect data provenance, model uncertainty, peer comparisons, and alert rationales, and if they retain authority to request additional information, challenge the ranking, or escalate a case for deeper review. Otherwise, supervisory attention may be allocated by an opaque score while human supervision remains only nominal.

Toward symbiotic decision intelligence

The endpoint of this discussion is not a larger checklist. It is a different view of human–AI hybrid financial intelligence. AI systems do not merely supply outputs to unchanged human workflows; they restructure attention, reshape what counts as evidence, and alter how organizations justify action. In this sense, human–AI hybrid finance involves *symbiotic decision intelligence*: a decision architecture in which human judgment, algorithmic processing, organizational routines, and governance controls jointly produce financial decisions.

This symbiotic view has three implications. First, cognition is restructured: analysts, underwriters, traders, claims adjusters, and compliance officers increasingly think with AI-generated summaries, rankings, alerts, and scenarios. Second, value is co-produced: financial decisions must balance predictive accuracy with risk appetite, fairness, customer treatment, regulatory expectations, and organizational legitimacy. Third, systems co-evolve with their environment: governed AI decisions change workflows, client behavior, market signals, and future data.

The mechanisms discussed in this section explain how human–AI hybrid financial decisions can be made more explainable, governable, and accountable. The next section extends the analysis from governance mechanisms to future decision architectures. Once AI is embedded in financial workflows, the future of human–AI hybrid finance is not only about better explanation or trust calibration. It is also about how institutions allocate authority over time, redesign work, deploy FinLLMs and agents, and monitor feedback loops in markets that respond to AI-mediated action.

The future of human–AI hybrid finance

“AI’s challenge: when human–AI collaboration works, and when it fails” and “Making human–AI hybrid finance work: explainability, trust, and governance” sections examined the conditions and governance mechanisms of human–AI hybrid finance. This section turns to future workflow and market-level implications. The future of AI in finance will be shaped less by isolated models than by *decision architectures*: workflows that allocate authority, structure overrides, govern bounded autonomy, and absorb feedback from customers, organizations, and markets. The relevant research question is therefore shifting from “Which model is most accurate?” to “Which hybrid design works best for which task, at which stage, under which governance constraints, and with what organizational and market feedback?”

This future should not be read as speculative. Much of the relevant capability already exists in partial form. LLMs can retrieve documents, summarize filings, answer finance questions, compare cases, draft reports, and interact with tools. Domain-specific systems such as BloombergGPT, FinGPT, FinanceBench, BizBench, FinBen, FinRobot, and InvestorBench show that the frontier has moved beyond narrow prediction toward financial reasoning, decision support, and agent-based evaluation (Wu et al. 2023; Yang et al. 2023; Islam et al. 2023; Krumdick et al. 2024; Xie et al. 2024; Yang et al. 2024; Li et al. 2025). Financial authorities have also begun to treat AI adoption as a financial stability and governance issue rather than a narrow model risk issue (Bank of England and Financial Conduct Authority 2024; Financial Stability Board 2024; OECD and Financial Stability Board 2024; Basel Committee on Banking Supervision 2024). Four issues are especially important: authority allocation, workflow redesign, FinLLMs and agents, and reflexive market feedback.

Decision architecture and authority allocation

Decision architecture can be described along three dimensions. The first is *stage*: information acquisition, interpretation, recommendation, approval, execution, monitoring, and learning. The second is *authority*: who has the right to act, override, escalate, or stop the system. The third is *accountability*: who is responsible for outcomes and for documenting the process. Many AI projects fail because these dimensions are not specified separately. They say the human remains responsible while the interface makes independent review difficult; or they say AI only recommends while defaults and thresholds make acceptance nearly automatic.

A useful supplement is to distinguish *initiation*, *intervention*, and *ratification*. Initiation asks whether humans or AI originate the candidate action. Intervention asks who can pause, contest, or modify that action. Ratification asks who has final authority. This framing is helpful because financial tasks vary not only in predictive difficulty but also in the reversibility of error, legal defensibility of action, and contestability of outcomes. When data trustworthiness is low, or when mistakes are difficult to reverse, more authority should remain with human validation and ratification. When tasks are routine, labels are abundant, errors are observable, and strong controls are in place, greater automation may be justified (Fügener et al. 2022; Gonzalez and Heidari 2025).

A practical test is the *counterfactual override question*: if the human disagrees with the AI, what happens? If override is impossible, AI is effectively deciding. If override is possible but costly, visible, or discouraged, authority is shared but asymmetric. If override is easy but never audited, the system may lose the benefits of standardization and learning. If override requires justification and later review, the organization can learn when humans add value and when they introduce noise or bias. This is why meaningful oversight, as defined in “[Making human–AI hybrid finance work: explainability, trust, and governance](#)” section, is not satisfied by a ceremonial human click. It requires that reviewers have access to evidence, understand uncertainty and constraints, have time and incentive to intervene, and can record reasons for acceptance, override, or escalation.

In future agentic workflows, the delegation frontier becomes more dynamic because AI systems may not only recommend actions, but also initiate tasks, call tools, execute bounded actions, and learn from outcomes. A bank may allow AI to acquire documents and generate a risk summary, require human approval for pricing and denial, automate low-risk monitoring, and escalate unusual cases. An asset manager may use an LLM to summarize filings, a predictive model to screen securities, a portfolio optimizer to propose weights, and a human committee to approve positions. A trading desk may automate execution but keep human control over strategy, risk limits, and kill switches. The key question is not whether finance should be human or automated in the abstract, but which stages should be human-led, AI-assisted, AI-led with oversight, or agentic.

Organizational adaptation and workflow redesign

GenAI and AI agents will not simply substitute for financial professionals; they will reshape workflows. Evidence from knowledge work shows that GenAI can improve productivity in writing, customer support, and professional tasks, but benefits depend on task fit and worker expertise (Noy and Zhang 2023; Brynjolfsson et al. 2025; Dell’acqua et al. 2026). In finance, this implies that the future unit of analysis is the *workflow* rather than the worker or the model alone. Analysts may spend less time collecting and summarizing documents and more time evaluating assumptions, scenarios, and strategic implications. Credit officers may become exception reviewers and relationship interpreters. Claims adjusters may review AI-generated damage estimates and coverage rationales. Compliance investigators may shift from screening every alert to auditing AI-generated triage and narratives. Risk managers may monitor model drift, third-party dependencies, and cross-model interactions rather than static validation reports (Cao et al. 2024; Eisfeldt and Schubert 2025).

This is more than labor substitution. It is a form of *cognitive re-bundling*: sensing, interpreting, reasoning, validating, documenting, and acting are redistributed across people, models, tools, and approval paths. Workflow redesign is also a *value-translation problem*. In credit approval, fairness, inclusion, profitability, and prudence may need to be negotiated across multiple checkpoints rather than compressed into a single score. In insurance, customer treatment, fraud control, claims efficiency, and policyholder contestability may need to be balanced across triage, adjustment, and communication. In capital markets, fiduciary duty, market integrity, and resilience under stress may need to be embedded in the workflow rather than treated as downstream compliance checks.

The organizational consequences are uneven. Junior analysts may benefit because AI reduces search costs, or they may lose training opportunities because routine analysis is automated. Senior professionals may become more productive, but their tacit knowledge may become harder to transfer. Compliance teams may process more cases, yet face new burdens of validating AI-generated narratives. Model risk teams may become bottlenecks if every workflow embeds multiple AI components. These changes affect incentives, career paths, and organizational learning. They also make the *jagged technological frontier* especially relevant: AI may be excellent at summarization and first drafts while weak at causal reasoning, strategic interpretation, or novel exception handling (Dell’acqua et al. 2026).

Future empirical work should therefore treat *process data* as at least as important as outcome data. To understand human–AI hybrid workflows, researchers need audit trails showing what recommendation was displayed, what evidence and uncertainty were visible, whether the human requested explanation, whether an override occurred, how long the review took, what justification was logged, and what downstream outcome followed. Such data make it possible to study organizational learning, the reliance wedge, and responsibility diffusion rather than only final decision accuracy.

GenAI, FinLLMs, and agentic finance

The most visible future frontier is the movement from *prediction* to *recommendation* and then to *agency*. Traditional financial AI mainly produced scores, forecasts, or rankings. FinLLMs and tool-using systems increasingly produce summaries, comparisons, plans, draft communications, and candidate actions. Benchmarks such as FinanceBench, BizBench, FinBen, FinRobot, and InvestorBench are valuable because they move beyond generic language understanding toward financial reasoning, grounded retrieval, decision support, and agent-based evaluation (Islam et al. 2023; Krumdick et al. 2024; Xie et al. 2024; Yang et al. 2024; Li et al. 2025).

Building on recent work on AI-agent autonomy and agentic finance, we propose a practical taxonomy of agentic finance that distinguishes five levels of capability (Feng et al. 2025; Gong 2026). Level 1 systems retrieve information and summarize sources. Level 2 systems analyze and compare alternatives, such as firms, loans, portfolios, claims, or alerts. Level 3 systems recommend actions and explain trade-offs. Level 4 systems execute bounded actions, such as preparing orders, routing cases, updating records, or sending draft communications after human approval. Level 5 systems adapt over time by planning multi-step workflows, coordinating tools, and learning from outcomes. This taxonomy is finance-specific, but it is consistent with broader arguments that agent autonomy should be treated as a design choice rather than an inevitable consequence of capability, and that the risks of AI agents increase as users cede greater control over external actions (Feng et al. 2025; Mitchell et al. 2025). Accordingly, most deployable financial systems in the near term should probably remain between Levels 1 and 4, with strong controls on any external action.

The central control variables for bounded agentic finance are *permissions*, *memory*, *grounding*, *approval*, and *logging*. Permissions define which tools the agent can use. Memory determines what the agent retains about prior interactions or clients. Grounding determines which data sources support its outputs. Approval determines when

humans must intervene. Logging determines whether the institution can reconstruct what happened. Without these controls, agents create accountability gaps. With them, agents can become useful workflow components rather than uncontrolled autonomous actors.

A bounded RegTech agent illustrates these controls. Such an agent might monitor regulatory updates, map new obligations to affected products or business units, draft a compliance gap analysis, and create tasks for control owners. However, it should not automatically revise policies, submit regulatory filings, or communicate externally without human approval. Permissions, grounding, approval gates, and action logs are therefore not peripheral safeguards; they define the boundary between useful compliance automation and uncontrolled regulatory action.

This shift also changes the meaning of a benchmark. A static benchmark asks whether a model answers a question correctly. An agent benchmark must ask whether the system selects the right tools, respects constraints, maintains state, asks for approval when needed, records its actions, and recovers from errors. Evaluation should therefore include *process correctness*, not merely final-answer correctness. It should test not only normal cases but also adversarial prompts, incomplete information, conflicting objectives, stale data, and unusual market conditions. More broadly, future work should study agentic finance in bounded and dynamic environments, not only on static one-shot tasks.

The most promising near-term applications are likely to be bounded rather than fully autonomous: document collection for credit files, first-pass research memos, compliance narrative drafting, portfolio monitoring, claims triage, client-question routing, regulatory-change mapping, and alert prioritization. These tasks are valuable because they are information-intensive and repetitive but still allow human verification. The highest-risk applications are those where agents can transact, communicate externally, alter records, or influence customers without adequate controls. The distinction between bounded assistance and autonomous action should therefore remain explicit in both research and policy.

Feedback loops, reflexivity, and the reflexive AI loop

The final frontier is *reflexivity*. AI in finance does not only analyze markets and institutions; it changes them. We use the term *reflexive AI loop* to describe the process through which AI-mediated decisions reshape the informational, behavioral, and market environment that future human and AI decisions will rely on. This idea connects performative prediction, strategic adaptation, and algorithmic monoculture (Perdomo et al. 2020; Hardt et al. 2016; Kleinberg and Raghavan 2021). Reflexivity should therefore be treated as endogenous rather than residual.

In lending, borrowers and platforms may adapt to scoring rules, documentation requirements, or AI-driven screening thresholds. In insurance, policyholders or intermediaries may adapt application narratives, claims documentation, or telematics behavior once they infer how AI-assisted underwriting or claims systems respond. In disclosure and capital markets, firms may increasingly write for machine readers, optimize textual structure for automated analysis, or adapt communication once they understand which signals move prices and recommendations. In trading and portfolio

management, common models, shared vendors, and similar data sources may amplify correlated positioning and crowding. In each case, local efficiency gains can come at the cost of broader distortions or fragility (Danielsson et al. 2022; Financial Stability Board 2024; Leitner et al. 2024).

Systemic risk supervision is also reflexive. If many institutions rely on similar third-party AI models for liquidity forecasting, credit monitoring, or portfolio risk dashboards, their responses to the same signals may become correlated: they may tighten credit, reduce exposures, rebalance portfolios, or hoard liquidity at the same time. Supervisory AI can create another feedback channel. Once institutions learn which patterns are likely to trigger supervisory attention, they may adapt reporting practices, internal controls, or disclosure language toward what they believe the supervisory model will flag. In this setting, AI changes not only individual decisions but also the behavior of supervised institutions and the data observed by supervisors.

These feedback effects operate at multiple levels. At the *individual* level, AI changes attention, interpretation, and action. At the *organizational* level, it changes workflows, documentation, and escalation. At the *market and system* level, it changes prices, liquidity, credit allocation, concentration, and stability. A central measurement problem follows: observed improvement may reflect genuine prediction gains, changed selection into actions, strategic adaptation by other actors, or redistributed externalities. This is why reflexive systems cannot be evaluated by historical validation alone. Once AI changes behavior, the data-generating process changes as well.

Future work should therefore combine historical validation with shadow deployment, post-deployment monitoring, simulations, and multi-level evidence. Institution-level studies can examine productivity, costs, fairness, and governance. Market-level studies can examine liquidity, volatility, concentration, and correlated behavior. Consumer-level studies can examine access, understanding, welfare, and recourse. Human–AI hybrid finance is not a one-time model-selection problem; it is an adaptive governance problem.

Conclusion: from prediction to delegation

This review has argued that “AI in finance” is now too narrow a framing. The literature has already shown that AI can improve prediction, classification, ranking, anomaly detection, retrieval, and document processing across banking, capital markets, asset management, insurance, risk, and compliance. The next question is how these capabilities are embedded in *human–AI hybrid financial decision systems*. Financial decision-making is increasingly hybrid because prediction is entangled with interpretation, recommendation, justification, execution, monitoring, and governance. The evidence also suggests that complementarity is not automatic. Human–AI systems can outperform either humans or algorithms alone, but only when complementarity is actively designed rather than assumed (Vaccaro et al. 2024; Fügener et al. 2022).

The central shift is therefore from *prediction* to *delegation*. Future research should ask not only whether AI can predict default, returns, fraud, sentiment, claims, or risk, but when AI should recommend, decide, execute, explain, learn, and be overridden. In that sense, human–AI hybrid finance is not merely the next application area of AI. It is an emerging theory of how financial decisions will be produced, supervised, and legitimized.

The introduction defined the core constructs as components of an integrated decision-system framework. Table 9 returns to them from the perspective of cumulative research, summarizing the core question and typical empirical use of each construct.

These constructs help reposition human–AI hybrid finance from a descriptive application area into a coherent research program. They also provide a positioning device for future studies. A paper can specify its financial setting, decision stage, AI function, human role, authority allocation, reliance mechanism, governance control, and outcome level. This makes it easier to compare apparently different studies—for example, LLM-assisted investment research and AI-supported credit underwriting—within a shared analytical language.

For theory, the review suggests that financial AI should be studied as a problem of *distributed decision rights*. Standard finance research often treats information as entering prices, portfolios, or contracts. Human–AI hybrid finance asks who or what converts information into action, under what constraints, and with what legitimacy. This implies three broader shifts. First, the unit of analysis should move from the isolated model to the human–AI hybrid financial decision system. Second, performance should be theorized as multidimensional rather than reducible to accuracy alone. Third, reflexivity should be modeled as central, not residual: once AI systems become common in lending, execution, disclosure analysis, claims, and regulation, they change the environment they are supposed to analyze.

For empirical research, future work should move beyond static benchmarks. Field experiments can test whether AI assistance improves real financial decisions. Shadow deployments can compare human-only, AI-only, and human–AI hybrid workflows without changing customer outcomes. Natural experiments can study staggered AI adoption across institutions or teams. Audit-log and process-mining studies can reveal how professionals actually accept, modify, or override AI advice. Laboratory and online experiments can isolate trust, uncertainty communication, and explanation effects using realistic financial tasks. Market-level simulations can study algorithmic monoculture, feedback loops, and systemic risk. Scientific progress will depend

Table 9 Core constructs for cumulative research on human–AI hybrid finance

Construct	Core question	Typical empirical use
Delegation frontier	At which stage should authority shift from humans to AI, remain human-led, or be shared?	Compare initiation, intervention, ratification, and override structures across lending, trading, advice, insurance, and compliance workflows
Reliance wedge	How far does realized human reliance depart from warranted reliance given system quality and incentives?	Study overreliance, underreliance, interface design, workload effects, and override behavior using audit logs and experiments
Decision-useful XAI	Do explanations improve actionability, contestability, auditability, and accountability rather than merely perceived transparency?	Evaluate explanations by whether they improve decisions, recourse, supervisory review, and governance outcomes
Meaningful oversight	Does human review remain substantively capable under speed, complexity, and organizational pressure?	Test whether reviewers have information, authority, time, independence, and responsibility in AI-mediated workflows
Reflexive AI loop	How do AI-mediated actions reshape organizational routines, customer behavior, disclosure practices, and markets?	Connect institution-level deployment to performativity, strategic adaptation, market feedback, and systemic externalities

on linking theory, platform infrastructure, and scenario validation rather than relying on one-shot benchmark gains alone.

Methodologically, human–AI hybrid finance needs evaluation dashboards rather than single metrics. Model-layer metrics such as accuracy, calibration, robustness, and fairness remain necessary. But they should be complemented by human–AI metrics such as reliance quality, override value, user understanding, and cognitive load; decision metrics such as expected utility, downside risk, and consumer welfare; workflow metrics such as speed, escalation quality, and audit completeness; governance metrics such as contestability, compliance, and accountability; and system metrics such as liquidity, volatility, concentration, and stability. For practice, the implication is equally clear: AI governance should be designed with the workflow, not appended after model development. Institutions should identify decision stages, assign authority, define escalation rules, select explanation formats, log reliance and overrides, monitor feedback, and distinguish between AI that informs, AI that recommends, AI that executes, and AI that learns.

Several limitations should be acknowledged. First, the empirical evidence on human–AI hybrid finance is uneven across domains. Credit scoring, trading, fraud detection, and AML have relatively mature bodies of AI research and deployment experience, whereas insurance, RegTech, systemic risk supervision, and agentic financial workflows remain less developed or more fragmented. Second, the governance implications of human–AI hybrid finance vary across regulatory jurisdictions. Requirements for explainability, human oversight, automated decision-making, consumer recourse, model risk management, and supervisory accountability differ across legal and institutional settings, so the framework should be applied with sensitivity to local regulatory context. Third, parts of the discussion are necessarily forward-looking. FinLLMs and agentic finance are evolving quickly, and many current evaluations are based on benchmarks, prototypes, sandbox settings, or early deployments rather than long-run field evidence. The framework proposed here should therefore be understood as a guide for organizing cumulative research and governance questions, not as a definitive assessment of the realized effects of all AI systems in finance.

A successful human–AI hybrid financial decision system is not one in which humans always agree with AI, nor one in which AI always dominates humans. It is one in which disagreement is informative, reliance is calibrated, authority is explicit, explanations are decision-useful, oversight is meaningful, and feedback is used to improve both model and workflow. The promise of AI in finance is not that machines will replace judgment everywhere. It is that financial decisions can be redesigned so that computational scale and human responsibility reinforce rather than undermine each other. The risk is that institutions adopt AI as a faster prediction engine while neglecting authority, reliance, explanation, oversight, and feedback. The most valuable future research will therefore study AI not as an isolated model, but as part of an evolving human–AI hybrid financial decision system.

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